

Piping Material Specification

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1. GENERAL

1.1. Scope

This Specification is applicable for the design and selection of piping materials during execution of Karabatan Power Plant Project.

1.2. Normative References

1.2.1. Any references of the following documents shall form a part of this specification. For documents with dates indicated, any amendments or revised versions thereafter shall not be applicable to this specification. For documents without indication of dates or number of modification (version), the latest version shall be applicable to this specification.

1.2.2. International Standards

API 594	Check Valves: Wafer, Wafer-Lug, and Double Flanged Type
API 598	Valve Inspection and Testing
API 599	Metal Plug Valves - Flanged and welding Ends
API 600	Bolted Bonnet Steel Gate Valves for Petroleum & Natural Gas Industries
API 602	Steel Gate, Globe and Check Valves for Sizes DN100 and Smaller for the Petroleum and Natural Gas Industries
API 603	Corrosion-Resistant, Bolted Bonnet Gate Valves--Flanged & Butt-Welding Ends
API 607	Fire Test for Soft-Seated Quarter-Turn Valves
API 608	Metal Ball Valves – Flanged, Threaded and Welding Ends
API 609	Butterfly Valves: Double Flanged, Lug and Wafer-Type
API 941	Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants
ASME B1.20.1	Pipe Threads, General Purpose (Inch)

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ASME B16.5	Pipe Flanges and Flanged Fittings (NPS 1/2 Through NPS 24)
ASME B16.9	Factory-Made Wrought Butt welding Fittings
ASME B16.10	Face-to-Face & End-to-End Dimensions of Valves
ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME B16.20	Metallic Gaskets for Pipe Flanges - Ring-joint, Spiral-wound, and Jacketed
ASME B16.21	Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.25	Buttwelding Ends
ASME B16.34	Valves—Flanged, Threaded, and Welding End
ASME B16.47	Large Diameter Steel Flanges (NPS 26 Through NPS 60)
ASME B16.48	Line Blanks
ASME B31.1	Power Piping
ASME B31.3	Process Piping
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASME B36.19M	Stainless Steel Pipe
MSS SP-43	Wrought Stainless Steel Butt-Welding Fittings
MSS SP-67	Butterfly Valves
MSS SP-68	High Pressure Butterfly Valves with Offset Design
MSS SP-70	Cast Iron Gate Valves Flanged and Threaded Ends
MSS SP-71	Gray Iron Swing Check Valves, Flanged and Threaded Ends
MSS SP-72	Ball Valves with Flanged or Butt-Welding Ends for General Service

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MSS SP-78	Cast Iron Plug Valves, Flanged and Threaded Ends
MSS SP-80	Bronze Gate, Globe, Angle and Check Valves
MSS SP-81	Stainless Steel, Bonnetless, Flanged Knife Gate Valves
MSS SP-83	Class 3000 Steel Pipe Unions Socket Welding and Threaded
MSS SP-85	Cast Iron Globe & Angle Valves Flanged and Threaded Ends
MSS SP-95	Swage(d) Nipples and Bull Plugs
MSS SP-97	Forged Integrally Reinforced Branch Outlet Fittings— Socket Welding, Threaded and Butt welding Ends
MSS SP-110	Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flanged Ends
MSS SP-118	Compact Steel Globe & Check Valves - Flanged, Flangeless, Threaded & Welding Ends (Chemical & Petroleum Refinery Service)
NACE MR0175	Petroleum and Natural Gas Industries- Materials for Use in H ₂ S-Containing Environments in Oil and Gas Production
BS 1868	Specification for Steel Check Valves (Flanged and Butt-Welding Ends) for the Petroleum, Petrochemical and Allied Industries
BS 1873	Specification for Steel Globe and Globe Stop and Check Valves (Flanged and Butt-Welding Ends) for the Petroleum, Petrochemical and Allied Industries
BS EN ISO17292	Metal Ball Valves for the Petroleum, Petrochemical and Allied Industries
ISO 10497	Testing of Valves -- Fire Type-Testing Requirements

1.2.3. Kazakhstan National Standards, Industrial Standards and Local Laws and Rules.

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SNiP PK 3.05-01-2010	Trunk Pipelines
SNiP RK 3.05.09-2002	Process Equipment and Process Pipelines
PB 03-585-03	Rules of Construction and Safety Operation of Technological Pipelines
ПБ 03-75-94	Safety Requirements for Steam and Hot Water Pipes
SNIP III-42-80	Main Pipelines

2. GENERAL REQUIREMENTS

2.1 All the piping shall be designed, manufactured, welded, heat treated (if required), inspected and tested as per the standards and specifications such as ASME B31.3 and B31.1 etc.

2.2 All the pipes and piping components in piping system shall be in conformity with the dimensions, materials and other requirements in the standards such as ASTM, ANSI/ASME, API, MSS and BS etc.

2.3 If the materials are not included in the above recognized standards but are in accordance with the standard of manufacturer, they shall have written approval by the Owner/PMC.

2.4 Alloys with low melting points such as aluminum, brass and bronze shall not be used with inflammable media or flammable liquid media. Aluminum can be used for the media with low temperatures.

2.5 The limits for use of copper and copper based alloys are as following:

2.5.1 Ammonia: the alloys with the copper content exceeding 50% cannot be used for ammonia media or ammonia liquor with PH value over 8.5.

2.5.2 Alkali: cannot use the alloys with copper content less than 85%.

2.5.3 Acetylene: the alloys with copper content over 50% cannot be used for the acetylene media existing in wet environment.

2.5.4 Copper cannot be used in the environment with sulfur or hydrogen sulfide.

2.6 The minimum weld coefficient of welded pipe shall be 0.85. For welded pipe, the following requirements apply:

2.6.1 Manufacturers shall perform hydro-test for all longitudinal or spiral welded pipes.

2.6.2 Manufacturers shall test 100 percent of weld for the electrical resistance welded pipes with ultrasonic method. Manufacturers shall perform 10 percent test for EFW pipes with radiograph. At least 250mm on each pipe end shall be radiographed.

In case pipe thickness exceeds 12.7mm, base metal (plate) shall be examined by magnetic (S6 of ASTM standard) or ultrasonic (S10 of ASTM standard) methods.

2.6.3 Furnace welded pipes are not permitted.

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2.6.4 Spiral welded pipes within process units are only used for water. Spiral welded pipes shall be double submerged welded. ERW pipes are only permitted for water services.

2.6.5 The peaking is limited to 3 mm (1/8 in) that is measured with Owner/PMC approved template method, when longitudinal welded pipes of CS and low alloy steel are used in the creep range (great than 750°F (400°C) for CS, great than 850°F (455°C) for alloy steel of C-1/2Mo and Cr-Mo), or used under severe cycling conditions (such as defined in ASME B31.3). Peaking is the deviation of the pipe outside contour from a true circle at longitudinally welded joints.

2.6.6 Pipes of alloy 1 1/4Cr-1/2Mo with longitudinal weld shall not be used in the conditions of design temperature greater than 900°F (480°C) unless exception to be submitted to Owner / PMC approval.

2.6.7 Design temperature for CS pipes with longitudinal weld shall not be higher than the following:

Electrical resistance welded pipe - 1000°F (540°C)(*1)

Fusion welded ASTM A671, ASTM A672 or ASTM A691 Classes 22, 32 or 42

Submerged-arc welded pipes, API SPEC 5L- 1000°F(540°C)(*1)

Fusion welded pipes with wall thickness less than 19 mm (3/4 in) of Class 13 of ASTM A671, ASTM A672 or ASTM A691

Fusion welded pipes with wall thickness greater than or equal to 19 mm (3/4 in) of Grade 23, 33 and 43 of ASTM A671, ASTM A672 or ASTM A691 - 750°F (400°C)

Spiral welded pipes API 5L - 200°F (95°C)

Notes: (*1) Owner/PMC approval is required when design temperature under long period is in the range of 800°F (427°C) and 1000°F (540°C).

2.7 Welding between kinds of metals (ferritic and non-ferritic) shall be avoided to the maximum extent possible. If two kinds of metals are welded together, 100% radiographic examination shall be made. The heat treatment of welds interesting two different metals shall be approved by the Owner/PMC.

2.8 Cast iron and spheroidal graphite cast iron pipes are limited to water media without freezing (except for boiler feed water, in conformity with Class D of ASME B31.3). They shall be limited to non pressure, non hazardous sewer service and to fire and other water services up to 95°C and a maximum pressure of 17barg. Large steam traps having bodies cast from these materials are acceptable for use up to 8 barg saturated steam service when steel or ductile iron materials are not readily available.

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3. PIPING

3.1. Piping

Dimensions shall comply with ASME B36.10M and ASME B36.19M, see 3.1.3 for equivalent imperial / DN Nominal sizes

3.1.1 NPS 3/8, 1¼, 2½, 3½, 5, 7 and NPS 9 shall not be used. However, in cases where the use of these sizes cannot be avoided, the required piping material, fittings, flanges, valves and gaskets shall be the same material as next larger sizes in the applicable piping class.

3.1.2 NPS 3/8 piping that is used for utility stations is permitted in applicable utility piping classes only.

The following is an equivalent pipe size table:

Imperial (inch)	DN (Metric mm)	Imperial (inch)	DN (Metric mm)
½"	15	20"	500
¾"	20	24"	600
1"	25	26"	650
2"	50	28"	700
3"	80	30"	750
4"	100	36"	900
6"	150	40"	1000
8"	200	44"	1100
10"	250	48"	1200
12"	300	52"	1300
14"	350	56"	1400
16"	400	60"	1500
18"	450	64"	1600

3.2. Flanges

3.2.1 Flange facing finish shall comply with ASME B16.5 – 6.4.5 and ASME B16.47 – 6.1.4

3.2.2 Unless otherwise noted Class 150 to Class 900 flanges in sizes that are beyond the scope of ASME B16.5 shall be in accordance with ASME B16.47 series A.

3.2.3 Class 150 FF flanges mating with ASME B16.1 (Class 125) flanges in sizes above NPS 24 shall be ASME B16.47 Series A (thro' NPS 60) or AWWA C207 Class D

3.2.4 Flanges not conforming to ASME B16.5, ASME B16.1, or ASME B16.47 shall be specified on piping drawings.

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3.2.5 Sizes for reducing or blind flanges are indicated by nominal pipe sizes.

3.2.6 Minimum size for orifice flanges is 1 1/2". Orifice unions are not permitted.

3.2.7 The bore of welding neck flanges shall correspond to the inside diameter of the connecting pipe or fitting, permitting a thickness difference up to 1.5 mm (1/16") When the difference in wall thickness between the two components being joined exceeds 1.5 mm (1/16"), taper boring shall be performed on the component having the heavier wall per ASME B31.3 (328.4.3 and 328.4.2).

3.2.8 Piping flanges not shown in the piping classes but that are required to mate to control valves, spec breaks at valves, equipment, or vessel nozzles, are permitted. (For example: Higher rating flanges are permitted in lesser rating piping system, or ASME B16.47 series A flanges are allowed to match equipment/vessel nozzles). All mating flanges with applicable line class bolting and gasket materials shall be provided as indicated on piping BOM.

3.3. Fittings

3.3.1 Reducing Fittings

3.3.1.1 Reducing fittings shall only be utilized as indicated in the individual Piping Class Branch Connection Chart, unless otherwise indicated on the P&ID or piping drawings.

3.3.1.2 Reducing tees, NPS 1½ and smaller, may be fabricated, from full size tees and reducing inserts. The maximum reduction for the utilization of reducing inserts is two sizes.

3.3.1.3 Reducers (concentric or eccentric) are the preferred fittings for all butt-weld end reductions.

3.3.1.4 Swage Nipples shall be used for reductions between butt-weld end and socket-weld or threaded end pipe and piping components. The maximum size of any swage shall be NPS 3.

3.3.1.5 Reducing couplings shall be used for line reductions NPS 1½ and smaller where the centerline of pipe elevations are to be maintained. Reducing couplings are preferred in lieu of swages for sizes NPS 1½ and less.

3.3.2 Threaded joints shall conform to American Standard taper (NPT) per ASME B1.20.1.

3.3.3 Forged steel threaded and socket weld fittings shall be per ASME B16.11.

3.3.4 Butt-weld fittings shall be in accordance with ASME B16.9. The bore of butt-weld fittings shall correspond to the inside diameter of the connecting pipe or piping components, permitting a thickness difference up to 1/16 in. When the difference in wall thickness between the two components being joined exceeds 1/16 in., taper boring shall be performed on the component having the heavier wall per ASME B31.3 (328.4.3 and 328.4.2)

3.3.5 NPS ¾ half-coupling is allowed only to be used to meet piping design requirement for drip legs or instrument details as shown on applicable drawings. The half-couplings shall have the same material and pressure rating as full-couplings specified in the applicable pipe classes.

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3.4. Gaskets

3.4.1 Unless otherwise noted non-metallic gaskets shall be 1.5 mm (1/16") thick after compression.

3.4.2 Unless otherwise noted spiral wound gaskets shall be 3.0 mm (C) thick after compression.

3.4.3 Kammprofile gaskets – Shall be considered as an alternative for class gaskets in sizes 1 ½" and below.

3.5. Bolting

3.5.1 Upon installation, standard bolt thread shall be exposed only on one side with the thread exposure of the extra bolt length to the other side of the flange. No stud bolt length shall be determined using approximately 1/3 of nut height as extra length equally spread on both sides of the flanged assembly.

3.5.2 Cap screws shall only be used when mating valves with threaded taps. These cap screws shall have the same material designation and grade as the stud bolts in the corresponding pipe classes. The valve manufacturer's stated tap depth and thread data shall be used as the basis for the final selection of cap screws.

3.5.3 Bolting 1 ½" diameter and larger in all services shall be suitable for tensioning. For critical, severe and / or high pressure services all diameter bolting below 1 ½" diameter shall in addition be reviewed for tensioning. Extra length bolts shall be used for tensioning and this extra length shall be protected after installation with metal/plastic caps

3.6. Vents and Drains

3.6.1 Valved and unvalved pipe vent and drain connections shall be NPS 1, unless otherwise noted on the engineering drawings. Engineering drawing is defined as P&ID, piping orthographic, piping isometric or referenced design standard. In case of conflict, the P&ID shall govern.

3.6.2 Hydrostatic vent and drain shall be the same components as unvalved vent and drain as indicated in modules in piping classes.

3.7. Threaded connections

3.7.1 Except for galvanized pipework and down stream of block valves threaded connections are not permitted.

3.7.2 Threaded plugs on hydrostatic vents, not provided with a valve, shall be seal welded after testing has been carried out.

3.7.3 All threaded connections that require seal welding, shall meet the requirements of the relevant fabrication standard.

3.7.4 Threaded connections, not requiring seal welding, shall be sealed as follows:

- a) 204°C & below- use PTFE Jointing Tape, or approved jointing compound.

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- b) Above 204°C - Seal weld wherever possible, otherwise use pipe jointing compound to manufacturer's standards.

3.8. Marking / tagging

3.8.1 All items shall be marked in accordance with Project requirements as stated on requisitions. Marking shall be in accordance with the relevant industry standard, plus any project specific information as required. All marking shall be clearly legible and not detrimental to the component.

3.8.2 Any items too small to be marked directly shall have permanently wired on stainless steel tags giving the required information. All wire shall be stainless steel.

3.9. Colour Coding

Refer to "Color Coding", 108184-0000-MA-SPC-0008.

3.10. Protection

3.10.1 After inspection and test, pipe, fittings and flanges shall be completely free of water, dried and prepared for shipment. Adequate protection shall be provided against mechanical damage and atmospheric corrosion in transit, and for at least six months outdoor storage at job site prior to installation.

3.10.2 Exposed finished and machined surfaces shall be given a heavy coating of Project approved rust inhibitor compound. Internal metal surfaces shall be sprayed or coated with a similar rust preventative. The rust inhibitor shall be easily removable in the field.

3.10.3 Weld bevels on pipe and piping components shall be clean and free of all foreign material before being coated inside and out for a distance of approximately 75 mm with a rust preventative compound that is not detrimental to welding.

3.10.4 Pipe, fittings and flanges shall be protected against damage and the ingress of water and foreign matter. Weld ends shall be protected by suitable caps, securely fastened, which will be suitable for site exposure as para 3.10.1 above. Flanged ends shall be protected by plastic (preferred) or wooden flange protectors which shall be held in place by a minimum of four bolts / inserts. End protection being offered should be clearly identified at the bid stage.

3.11. Y-Strainers

3.11.1 Unless otherwise noted, Y-Strainers shown shall be furnished with heavy gage perforated 304SS screen (1/32" perforation up to NPS 2 and 1/16" perforation for NPS 3 & larger sizes) and shall comply with line class material/pressure-rating per ASME B16.5 and B31.3 as per respective piping classes.

3.11.2 Y-Strainers for acid and caustic services or galvanized spec shall be furnished with flanged blowdown connections.

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3.12. T-Strainers

3.12.1 T-Strainers in sizes NPS 3 through NPS 24 shall have minimum 11 gage perforated stainless steel basket (with reinforcement rings where required) and shall comply with fitting material and wall schedule rating per ASME B16.9 as per respective piping classes. The in-line end connections of T-Strainers shall be butt-weld ends unless flanged ends are required by piping design as indicated on piping drawings.

3.12.2 T-Strainers for acid and caustic services or galvanized spec shall be furnished with flanged blowdown connections.

3.13. Chemical Composition

3.13.1 All carbon and low temperature carbon steel grades of material for pipe, fittings and flanges shall have a maximum carbon content of 0.25% and a maximum carbon equivalent of 0.45% based on the following formula:

$$CE = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}$$

% S ≤ 0.025

% P ≤ 0.025

3.14. Certification

3.14.1 All material shall be supplied with a minimum of BS EN 10204 type 3.1 certification, unless specifically noted differently on requisition and / or data sheets.

3.15. Impact Testing

3.15.1 Impact testing to be in line with ASME B31.3 §323.3 requirements as minimum.

4. VALVE

4.1. General

4.1.1 Unless otherwise specified, steel valves shall conform to ASME B16.34. However, the requirements of the plant design code (ASME B31.1, ASME B31.3 etc.) shall be taken into account as stated on the valve data sheet.

4.1.2 Face-to-face dimensions of steel flanged valves shall conform to ASME B16.10 to the extent covered. For valves not covered in ASME, certified dimensions drawings shall be provided by the Selected Vendor.

4.1.3 Unless otherwise noted, control valves should be flanged ends according to ASME B16.5 or B16.47A. The designer shall check instrument vendor data / drawings to verify flange connections and dimensions.

4.1.4 Isolation valves installed in relief systems shall meet the requirements of ASME B31.3 Section 322 (See paragraph 4.2.4 below)

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4.1.5 When permitted in piping class, wafer butterfly valves shall be used wherever possible. Butterfly valves that are threaded lug or dual flange design shall only be used as required for “dead end” service and these valves shall be clearly noted on the applicable piping drawing.

4.1.6 Ball, Butterfly and Plug valves shall be reviewed for cap screw requirements.

4.1.7 All valves 2” NPS and above shall be provided with a locking facility for the ‘open’ and ‘closed’ positions. The facility shall enable a padlock (supplied by others) with a minimum diameter of 5 mm to be fitted.

4.1.8 All valves shall be protective coated in accordance with project specifications. Alternative (higher grade vendor’s standard) coatings may be accepted if highlighted at bidding stage.

4.1.9 Any exposed (unpainted) carbon steel parts / surfaces shall be given a heavy coating of project approved rust inhibitor. Internal metal surfaces shall be sprayed or coated with similar suitable rust preventative.

4.1.10 Valve ends shall be suitably protected against damage and the ingress of foreign matter. Ends shall be protected as below:

4.1.10.1 Socketweld and plain ends shall be closed with metal or plastic protectors that fit either inside or outside the valve ends.

4.1.10.2 Threaded ends shall be fitted with plastic plugs / caps.

4.1.10.3 Flanged ends shall be protected by plastic (preferred), or wood covers, securely held in place by a minimum of four bolts / inserts. Gasket contact surfaces shall be protected to ensure serrations are not damaged in transit or storage.

4.1.11 All valves shall be marked / tagged with (as a minimum) the project tag and commodity code number in addition to information required by the applicable design standard. Full marking requirements will be stated within the enquiry.

4.1.12 All valves having a preferred direction of flow shall be marked with an arrow cast on the valve body.

4.1.13 Stamping or engraving shall not infringe on the minimum required wall thickness of the valve.

4.1.14 All flanged valves shall have end flanges conforming dimensionally to ASME B16.5 or 16.47A. Flange surface finishes will be stated on valve data sheets and will refer to the respective standard.

4.1.15 End and bonnet flanges shall be cast or forged integrally with the valve body. Conversion of weld end valves to flanged by welding on flanges is not permitted.

4.1.16 All valves shall be supplied with a minimum of BS EN 10204 type 3.1b certification. The level of certification will be stated on the valve data sheet.

4.1.17 Fugitive Emissions

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4.1.17.1 The valve manufacturer shall provide a written guarantee (warranty) stating that all steel valves are provided with 'low emission' bonnet gaskets, stem seals and packing. The detectable leakage fugitive emission rate shall be less than 500 ppm, but may be required to be less than this for critical applications. The valve data sheet will indicate the level of emission allowable.

4.1.17.2 Production tests are not required. However, all valves shall be capable of meeting the stated emission limits when and if tested in accordance with method 21 appendix "A" of 40 CFR60 and measured by an organic vapour analyser approved by purchaser.

4.1.18 Records of successful lab test results confirming the above shall be available for review by the Purchaser during quotation stages.

4.1.19 Valves specified as needing to be 'fire safe' shall have been tested and certified as having met the requirements of API 607.

4.1.20 When specified on valve data sheets any valve with weld deposits shall be impact tested as per project requirements. The testing will be for weld deposit, heat affected zone (HAZ) and Charpy impact test results will meet the minimum requirement for the project. The testing may be carried out on representative materials subject to Purchasers approval.

4.1.21 When stated on data sheets valves shall be suitable for full vacuum service. Long stuffing box shall be used. Vendor shall provide a written guarantee (warranty) to this effect.

4.1.22 Written records confirming the above shall be available for review by the Purchaser during the bidding stage.

4.1.23 Any valve stated as being in 'hydrogen service' on the valve data sheet shall have bonnet gasket and stem packing designed specifically for the application and shall provide zero leakage as stated in the Project specification. All valves shall be Helium tested.

4.1.24 Any valve stated as being in 'sour service' shall meet the requirements for sour environments as stated in NACE MR-01-75.

4.1.25 Vendor shall not modify an existing valve to bring into conformance with data sheet requirements without written approval of the Purchaser. Examples of modifications are as follows:

Addition of extended bonnet (gas column) to a standard valve.

Inserting resilient seats.

Revising trim.

Modifying flanged end valves to butt weld end (machining off flanges).

Changing bolting and / or packing.

Weld repair to castings and any subsequent heat treatment.

All weld repairs on heat treated casting shall be followed by a new heat treatment.

Modifying a butt weld end valve to threaded or socket weld ends.

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Modifying a threaded end valve to socketweld ends.

4.1.26 Buttweld, socketweld and threaded end carbon steel valves shall have a 0.25% maximum carbon content. Free machining steel shall not be used.

4.1.27 Unless otherwise specified, all austenitic stainless steel materials (300 series) shall be furnished in a solution annealed condition. Approved repairs, if any, shall be made prior to solution annealing.

4.1.28 Where a valve data sheet calls for body plugs, these shall be as a minimum equal in composition to the valve body. Valves shall not have drilled and plugged holes unless specifically requested on data sheets.

4.1.29 Where 'valve trim' is listed on valve data sheet, this shall mean all wetted internal parts of the valve.

4.1.30 Bonnet gaskets shall be Manufacturer's standard (non-asbestos), unless stated otherwise on the valve data sheet. Gasket shall be fully detailed in quotation.

4.1.31 Metallic and non-metallic bonnet and cover gaskets shall have corrosion resistance equal (or better) than that of the body and bonnet materials.

4.1.32 Rings for valves with ring-joint type bonnet or cover shall have a maximum hardness in accordance with the following:

Soft Iron	90 BHN
5% Chrome	130 BHN
Austenitic Stainless Steel	160 BHN
Monel	125 BHN

4.1.33 The Purchaser shall provide documentation of examination, inspection and testing as specified in the relevant ASME standard specification. All test documentation shall be certified 'true' by the Vendor.

4.1.34 Each individual valve shall be subjected to a shell and seat pressure test, at the pressures stated on the valve data sheet.

4.1.35 Any optional backseat, low pressure closure or high pressure closure test shall be clearly identified on the valve data sheet (with relevant test pressure / temperature). Vendor shall include all additional costs for these tests.

4.1.36 All pressure testing of austenitic stainless steel valves shall be carried out with water having a maximum of 50 ppm chloride content.

4.2. Gate and Globe Valves

4.2.1 Unless otherwise noted, OS & Y valves shall be supplied with bevel gear operators for the following class and size of valve:

ASME Class	Gate Valves	Globe Valves
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150	NPS 16 and larger	NPS 14 and larger
300	NPS 12 and larger	NPS 10 and larger
600	NPS 10 and larger	NPS 8 and larger
900	NPS 8 and larger	NPS 6 and larger
1500	NPS 6 and larger	NPS 6 and larger
2500	NPS 6 and larger	NPS 4 and larger

4.2.2 Gate Valves, Double Seated Design for Steam Service

4.2.2.1 The location and orientation of double seated gate valves, where specified, in ASME Class 600 and greater and in sizes NPS 20 and larger shall be reviewed (process systems/piping Eng.) for various operation modes to minimize the potential for trapping of liquid in the center cavity.

4.2.3 Gate valves NPS 11/2 and smaller shall be conventional port, unless otherwise noted.

4.2.4 Gate valves used as isolation valves in pressure relief systems should be installed with stems oriented horizontally or if this is not feasible the stem could be oriented downward to a maximum of 45 degree from the horizontal.(For the special case of low temperature applications individual review shall be performed by process systems/piping Eng.)

4.3. Quarter turn Valves (Butterfly, Plug, Ball)

4.3.1 Unless otherwise noted, quarter turn valves shall be supplied with worm gear operators as follows:

ASME Class	Plug Valves	Butterfly Valves	Ball Valves
150	NPS 6 and larger	NPS 8 and larger	NPS 6 and larger
300	NPS 6 and larger	NPS 6 and larger	NPS 6 and larger
600		NPS 4 and larger	NPS 4 and larger

4.3.2 Stems of ball valves shall be 'blow out proof' (contained within the valve body casting / forging), and shall be furnished with an anti-static device.

4.3.3 The operating levers (or handles) on ball valves shall indicate, by their position, whether the valve is open or closed. The open position shall be when the lever (or handle) points in a direction parallel to the flow through the valve. It shall also be impossible to re-assemble any valve to affect this indication.

4.3.4 All ball valves shall have self relieving cavity mechanisms. Details of this shall be supplied with bid. For seat supported (floating ball) type valves this may be achieved by use of a hole drilled in the ball subject to approval from purchaser. The use of a hole in ball will make the valve uni-directional.

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4.3.5 All wrench operated ball valves in insulated lines shall have an extended spindle (measured from the valve bonnet) to ensure that the wrench clears valve insulation. The spindle to have an outer stationary tube welded to the valve bonnet and having a weather seal at the wrench end. The stationary tube enables the insulation to be properly sealed around it.

- a) 100mm for valves 50mm NS and smaller.
- b) 150mm for valves 75mm NS and larger.

4.4. Check Valve

4.4.1 Lift check valves shall be installed horizontally only. Unless otherwise noted, all other types of check valves shall be installed horizontally or vertically with upward flow.

4.5. Blowoff Valves

4.5.1 The straightway (Y-type) or angle type blowoff valves may be used in vertical runs of pipe or they may be used in horizontal runs of pipe provided they are so constructed or installed that the lowest edge of the opening through the seat is at least 25 percent of the inside diameter below the centerline of the valve.

4.6. Plug Valves

4.6.1 Lubricated plug valves shall not be used where lubricant could contaminate process fluid and shall be used only when specified by the piping class or indicated on the P&ID.

4.7. Extended Bonnet Lengths / Low Temperature Service

4.7.1 Extended bonnet valves for low temperature service shall have the minimum extended bonnet lengths stated in the table below, measured from the centerline of the valve bore to the bottom of the packing (stuffing box).

Valve	Centerline of Valve Bore to Bottom of Valve Packing (mm)	
Size (NPS)	Rising Stem (i.e. Gate, Globe)	Quarter-Turn (e.g. Butterfly, Ball)
½, 3/4, 1	305	190
1½	355	216
2	406	254
3	457	330
4	558	355
6	610	432
8	685	457
10	889	635
12	1016	711

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14	1118	787
16	1194	838
18	1295	889
20	1397	940
24	1600	990

Centerline of valve bore to bottom of valve packing for valve sizes beyond those shown in the above Table require review and approval.

4.7.2 Only valves with an operating temperature of below minus 46° (as stated on data sheet) shall be furnished with extended stems.

4.7.3 In addition to point 4.7.2 above, consideration should be given to valves that are in services with an operating temperature of below minus 46° but are not required to be operated at these temperatures. In such cases valves should be able to 'seal and survive', which means having sealing components that can withstand the low temperatures and will recover to hold pressure as the temperature rises.

5. BRANCH CONNECTION

5.1 Branch connections for header sizes two inches and larger which require reinforcement other than sockolet or weldolet shall be in accordance with the applicable branch connection table and ASME B31.3, Figure 328.5.4D.

5.1.1 Reinforcement pad shall primarily be the same material as pipe. Plate material with compatible strength, weldability, heat treatment requirements, galvanic corrosion, thermal expansion etc. may be used as an alternative reinforcing pad material subject to contractors review and approval.

5.1.2 Thickness of reinforcement pad shall be the same as header wall thickness and pad width shall be two times the outside diameter of the branch pipe, unless otherwise specified by the stress analyst and shown on the piping isometric.

5.1.3 The need for a reinforcement pad or fitting could be eliminated on case by case basis.

5.1.4 Minimum spacing of branch connections shall be as follows: the clear distance between the edges of attachment welds for adjacent branch connections or reinforcement pads shall not be less than four times the thickness of the run wall, or 1 in. (25 mm), whichever is greater.

5.1.5 Pad reinforcement branch connections are not permitted whenever the piping design temperature is 800°F (427°C) or greater.

5.2 Substitutions may be made for stub-in or other fittings as calculated and determined by the stress engineer when high stress concentrations or any other reasons preclude the use of branch connection shown in the branch table. The branch connection as marked on the piping isometric as determined by the stress engineer shall override the branch table.

5.3 Integrally reinforced fittings may also be substituted for stub-in, reinforcement pads, or other fittings as determined by stress analyst when high thermal cycling or constant vibration

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is present in the piping system or high stress concentrations or any other reasons preclude the use of stub-in, reinforcement pads, or other fittings shown in the branch table.

5.4 Unless otherwise noted, the 45 degree lateral branch connections shall be as follows:

5.4.1 For 12 in. and smaller, use "Latrolet" type fitting.

5.4.2 For sizes larger than 12 in., consult with piping engineering.

5.5 The branch connections at elbows shall be as follows:

5.5.1 For 12 in. and smaller, use "Elbolet" type fitting.

5.5.2 For sizes larger than 12 in., consult with piping engineering.

5.6 On any of following occasions, the integral reinforcement branch welding fitting (O-let) with size $\geq 100\text{mm}$ (NPS 4) are not allowed to be welded to the outside surface of main pipes without Owner permission.

5.6.1 Ratio of d/D (diameter of branch pipe/ diameter of main pipe) is more than 0.8;

5.6.2 Wall thickness of main pipes is less than standard wall thickness (STD);

5.6.3 Outside diameter is more than 900mm (36 inch) while wall thickness of main pipes is less than 19mm (0.75 inch).

5.7 Unless otherwise noted or as previously indicated, 90° branch connections shall be made in accordance with the branch tables as indicated in the individual piping classes.

5.8 Unless otherwise noted, stub-in branch connection with or without reinforcement pad shall be sit-in type as shown in detail "2" or "4" of Fig. 328.5.4D of ASME B31.3.

5.9 Branch connections for vent, drain and sample connections, and for piping/instrument connections shall be based on the modules referenced in each pipe class.

5.10 Branch reinforcement requirements shall be in accordance with ASME B31.3, §304.3.2/3.